

- Differentiate between chemisorption and physisorption. (*Frequently Asked*)
- Discuss Freundlich adsorption isotherm and write its limitations.
- Discuss kinetics of enzyme catalysis and derive Michaelis-Menten equation.
- How many times the rate of reaction increases at 20°C for a reaction having the activation energies in the presence and absence of a catalyst as 50 KJ mol⁻¹ and 75 KJ mol⁻¹?
- Define rate constant. What is the unit of second order rate constant?
- Define adsorption isotherm. Derive Langmuir adsorption isotherm equation for adsorption of gaseous molecule on solid surface.
- Calculate the temperature for a reaction at which its half-life is 2 minutes. Given: $A = 5000 \times 10^{10} \text{ sec}^{-1}$ and $E_a = 10^2 \text{ Jmol}^{-1}$.
- Find out correct rate expression for the following reaction: $X \leftrightarrow Y$ (*Fast*) ; $2X + Y \rightarrow P$ (*Slow*).
- Explain the methods to determine V_{max} and K_m in Michaelis-Menten equation for enzyme catalysis.
- Differentiate between adsorption and absorption.
- Find correct rate law expression for following reaction: $A \leftrightarrow B$ (*fast*) ; $2A + B \rightarrow P$ (*slow*).
- Explain Ostwald's isolation method for determining order of a given reaction.
- Find correct rate law expression for following reaction: $A \leftrightarrow B$ (*fast*) ; $C \leftrightarrow D$ (*fast*) ; $2A + B + D \rightarrow P$ (*slow*).
- Explain differential or graphical method for determining order of a given reaction.
- Derive rate constant for second order reaction when both the reactants are different.
- Explain half life time method for determining order of a given reaction.
- Explain the differential rate expression method to determine the order of reaction (n).
- In a second order reaction the initial concentration of the reactant is 0.2 moles/L. The reaction is found to be 30% complete in 50 min. Calculate the value of rate constant.

Unit 3: Electrochemistry, Nanochemistry & Environmental Chemistry

- Calculate the potential of the following electrochemical cell at 25°C: $\text{Cu}_{(s)} / \text{Cu}^{2+}_{(aq)} (0.50\text{M}) \parallel \text{H}^+ (0.01\text{M}) / \text{H}_2 (0.95\text{atm}), \text{Pt}$. (Given $E^\circ_{\text{cathode}} = 0.00\text{V}$ and $E^\circ_{\text{anode}} = 0.34\text{V}$).